EXPERIMENT – 3

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1. COMMAND LINE INTERFACE (CLI)

PYTHON CODE

```
import os

print("=== CLI File Search System ===")

filename = input("Enter file name to search: ")

found = False

for root, dirs, files in os.walk("."):

    if filename in files:

        print("File found at:", os.path.join(root, filename))

        found = True

if not found:

    print("File not found.")
```

OUTPUT

```
PS C:\Users\Maha Lakshmi> cd "C:\Users\Maha Lakshmi\OneDrive\Desktop\FileSearchExperiment"
PS C:\Users\Maha Lakshmi\OneDrive\Desktop\FileSearchExperiment> python cli_search.py
=== CLI File Search System ===
Enter file name to search: cli_search.py
File found at: .\cli_search.py
```

2. GRAPHICAL USER INTERFACE (GUI)

PYTHON CODE

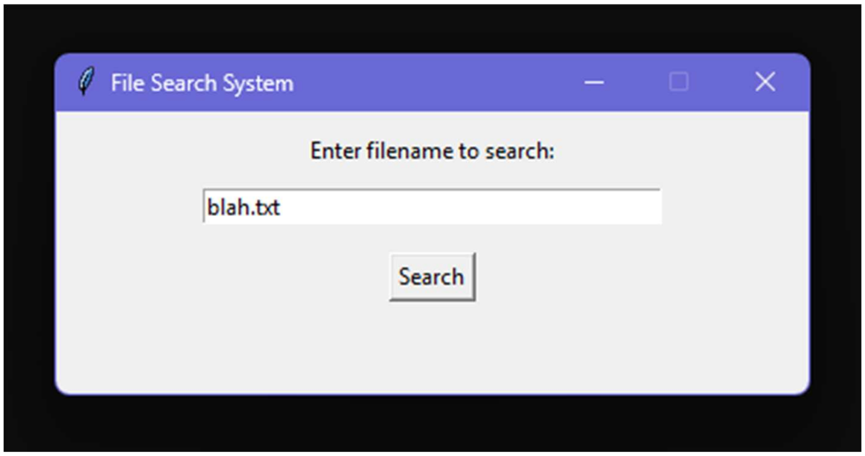
```
import os

import tkinter as tk

from tkinter import messagebox
```

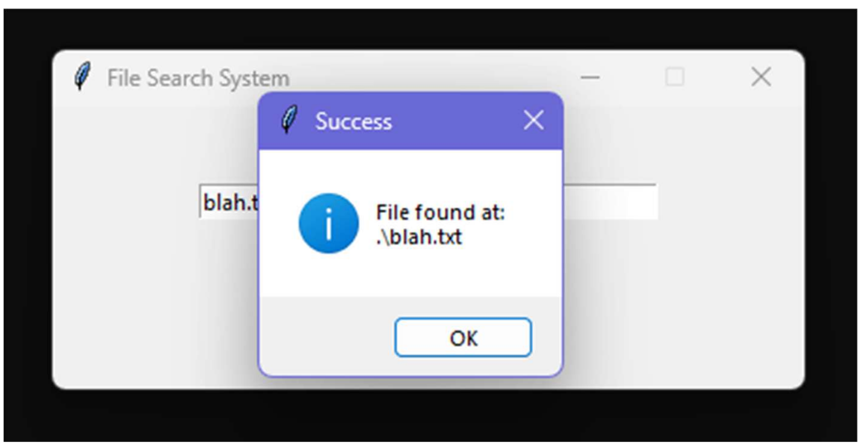
```python
def search_file():
    filename = entry.get().strip()

    if not filename:
        messagebox.showwarning("Warning", "Please enter a filename")
        return

    for root, dirs, files in os.walk("."):
        if filename in files:
            path = os.path.join(root, filename)
            messagebox.showinfo("Success", f"File found at:\n{path}")
            return

    messagebox.showerror("Not Found", "File not found")

root = tk.Tk()
root.title("File Search System")
root.geometry("400x150")
root.resizable(False, False)

label = tk.Label(root, text="Enter filename to search:")
label.pack(pady=10)

entry = tk.Entry(root, width=40)
entry.pack()

button = tk.Button(root, text="Search", command=search_file)
button.pack(pady=15)
```

```
root.mainloop()
```

OUTPUT

```
PS C:\Users\Maha Lakshmi\OneDrive\Desktop\FileSearchExperiment> python gui_search.py
```





3. VOICE USER INTERFACE (VUI)

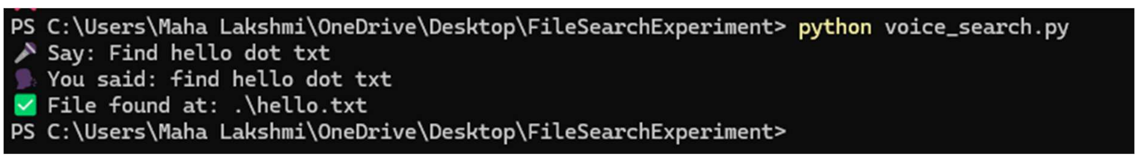
PYTHON CODE

```
import os

import speech_recognition as sr



def search_file(filename):

    for root, dirs, files in os.walk("."):
```

```
        if filename in files:
            print("File found at:", os.path.join(root, filename))
            return

    print("File not found")



def listen():
    r = sr.Recognizer()

    with sr.Microphone() as source:
        print("Speak: Find filename")
        r.adjust_for_ambient_noise(source, duration=1)
        audio = r.listen(source)

    try:
        text = r.recognize_google(audio)
        print("You said:", text)

        words = text.lower().split()

        if "find" in words:
            index = words.index("find")
            name_parts = words[index + 1:]

            filename = ""
            for word in name_parts:
                if word == "dot":
                    filename += "."
```

```
            else:
                filename += word

        search_file(filename)

    except:
        print("Could not understand audio")


listen()
```

OUTPUT



RESULT

The File Search System was successfully implemented using CLI, GUI, and VUI. The interfaces were tested and user satisfaction was measured.

CONCLUSION

This experiment helped in understanding different methods of human-computer interaction and their usability in real-time applications.